THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Coordinates** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circle has diameter with endpoints $(-3, 4)$ and $(21, 74)$. Its radius is: 1

(a) 38 **(b)** 37 **(c)** 36 **(d)** 39

2. The point on the y-axis equidistant from $(6, -4)$ and $(-6, -2)$ is: 1

(a) $(1, -3)$ **(b)** $(0, 3)$ **(c)** $(0, -3)$ **(d)** $(-3, 0)$

3. If the distance between $(x, 1)$ and $(-2, 13)$ is 13 and $x > -2$, then $x =$ 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 2

4. The fourth vertex of the parallelogram with consecutive vertices $(-2, 3)$, $(2, 1)$, $(5, 4)$ (in order) is: 1

(a) $(1, 6)$ **(b)** $(-1, 6)$ **(c)** $(1, -6)$ **(d)** $(6, 1)$

5. A point $P(x, 0)$ on the x-axis is 15 units from $(1, 12)$ with $x > 1$. The value of x is: 1

(a) 10 **(b)** 9 **(c)** 12 **(d)** 11

6. The points $(2, 3)$, $(2, 8)$ and $(2, -1)$ are: 1

(a) collinear, on a vertical line **(b)** collinear, on a horizontal line
(c) vertices of a triangle **(d)** vertices of a rectangle

7. The area of the triangle with vertices $(0, 0)$, $(13, 0)$ and $(0, 6)$ is: 1

(a) 38 **(b)** 40 **(c)** 39 **(d)** 41

8. The distance of the circumcentre of the right triangle with vertices $(0, 0)$, $(16, 0)$ and $(0, 30)$ from the origin is: 1

(a) 18 **(b)** 16 **(c)** 19 **(d)** 17

9. **Assertion (A):** The point $(4, -2)$ lies in Quadrant IV. 1
Reason (R): In Quadrant IV both the abscissa and the ordinate are negative.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The points $(2, 3)$ and $(3, 2)$ represent the same point on the Cartesian plane. 1
Reason (R): The order of the numbers in an ordered pair matters.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The end points of a diameter of a circle are $A(-4, 2)$ and $B(14, 26)$. Find the centre and the radius. 2

12. Find the distance between the points $(5, -9)$ and $(-1, 2)$. 2

13. The vertices of a right triangle are $O(0, 0)$, $A(16, 0)$ and $B(0, 30)$. Find the circumcentre and the circumradius. 2

14. The end points of a diameter of a circle are $A(0, -5)$ and $B(40, 37)$. Find the centre and the radius. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the points on the x-axis whose distance from $A(4, 40)$ is 41 units. 3

16. Three vertices of a rectangle $ABCD$ are $A(-2, 1)$, $B(16, 25)$ and $D(-34, 25)$. Find C and the diagonal AC. 3

17. Find the length of the median AD of the triangle with $A(0, 0)$, $B(22, 0)$, $C(0, 120)$, where D is the midpoint of BC. 3

OR

Find the points on the x-axis whose distance from $A(0, 35)$ is 37 units. 3

18. Three vertices of a rectangle $ABCD$ are $A(2, -3)$, $B(17, 17)$ and $D(-46, 33)$. Find C and the diagonal AC. 3

OR

Find the length of the median AD of the triangle with $A(0, 0)$, $B(28, 0)$, $C(0, 96)$, where D is the midpoint of BC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a treasure-hunt game on a coordinate grid, Imran buries a flag at F$(4, -3)$. His friend Tanvir must find the flag using reflections of this point. 5

(i) In which quadrant is the flag?
(ii) Find the reflection of F in the y-axis.
(iii) Find the reflection of F in the x-axis.
(iv) Find the point F′ obtained by reflecting F in both axes, and the distance FF′.

20. Show that $A(4, 0)$, $B(11, 24)$, $C(-13, 31)$, $D(-20, 7)$ are the vertices of a square. Find its side, diagonal squared and area. 5

OR

The point $P(x, 0)$ is equidistant from $A(-15, 24)$ and $B(-1, -24)$. Find P and the distance PA. 5

END OF QUESTION PAPER — Total: 40 Marks